

Practical (OS)

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1. Server code and output:

Code:

```
GNU nano 8.7.1 server.c
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int server_fd, client_fd;
    struct sockaddr_un address;
    char buffer[100];

    server_fd = socket(AF_UNIX, SOCK_STREAM, 0);
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);
    unlink(SOCKET_PATH);
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));
    listen(server_fd, 5);
    printf("Server waiting...\n");
    client_fd = accept(server_fd, NULL, NULL);
    read(client_fd, buffer, sizeof(buffer));

    printf("Message from client: %s\n", buffer);

    close(client_fd);
    close(server_fd);

    unlink(SOCKET_PATH);

    return 0;
}
```

Output:

```
gayathri@Gayathri:~$ nano server.c
gayathri@Gayathri:~$ gcc server.c -o server
gayathri@Gayathri:~$ ./server
Server waiting...
Message from client: Hello Server
gayathri@Gayathri:~$ |
```

2.Client Code:

```
GNU nano 8.7.1 client.c
#include <stdio.h>
#include <unistd.h>
#include <sys/socket.h>
#include <sys/un.h>
#include <string.h>

#define SOCKET_PATH "/tmp/my_socket"

int main()
{
    int client_fd;
    struct sockaddr_un address;
    client_fd = socket(AF_UNIX, SOCK_STREAM, 0);
    address.sun_family = AF_UNIX;
    strcpy(address.sun_path, SOCKET_PATH);
    connect(client_fd, (struct sockaddr *)&address, sizeof(address));
    write(client_fd, "Hello Server", 12);
    printf("Message sent to server.\n");
    close(client_fd);
    return 0;
}
```

Output:

```
gayathri@Gayathri: ~$ nano client.c
gayathri@Gayathri:~$ gcc client.c -o client
gayathri@Gayathri:~$ ./client
Message sent to server.
gayathri@Gayathri:~$ |
```